

Standard Operating Procedure: Pre-Assembled Fourier Ptychographic Microscopy (FPM) System Operation and Data Acquisition

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Abstract

This report details the operational, calibration, and data acquisition procedures for a pre-assembled, low-cost Fourier Ptychographic Microscopy (FPM) system. The aim is to build a low cost, miniature FPM setup with a Raspberry Pi camera and perform high-quality reconstructions. The guide covers system initialization, software calibration, standard scanning workflows, and data interpretation for computational high-resolution image synthesis.

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1 Phase 1: System Overview and Principles

1.1 Executive Summary

Fourier Ptychographic Microscopy (FPM) is a computational imaging technique that bypasses the traditional optical trade-off between field-of-view (FOV) and spatial resolution. By systematically illuminating a sample from varying angles using an LED array and capturing the resulting low-resolution diffraction patterns, phase-retrieval algorithms are used to computationally reconstruct an image with both a massive FOV and high resolution.

A primary challenge of this project is that the system needs careful calibration in order to have good quality reconstructions.

1.2 Pre-Assembled Hardware Components

The current FPM station is fully built and consists of:

- **Central Controller:** Raspberry Pi 3 board (acting as both hardware trigger and storage).
- **Illumination Source:** Pimoroni Unicorn HAT HD (16x16 RGB LED matrix) mounted directly to the Pi's GPIO pins.
- **Detector:** Raspberry Pi Camera Module, mounted facing downward.
- **Sample Manipulation:** Focusing stage and Sample platform.
- **Optomechanical Chassis:** A rigid 3D-printed fixture securing the camera at a fixed distance above the LED matrix, mounted on an optical breadboard.

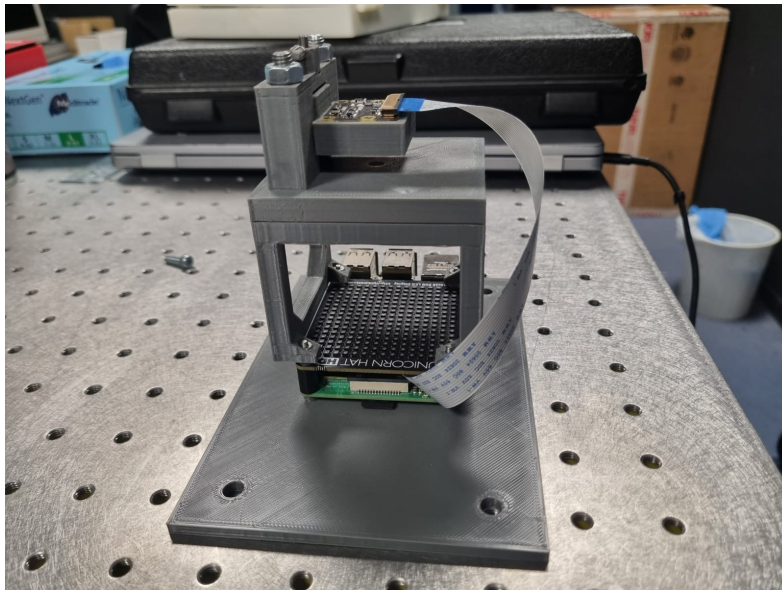


Figure 1: Side profile of the pre-assembled FPM station.

2 Phase 2: Pre-Operation Initialization & Calibration

Since the hardware is already assembled, this phase ensures the system is software-ready and optically aligned for the current session.

2.1 Flashing the Raspberry Pi OS

If the Raspberry Pi requires a fresh operating system installation or reset, follow these steps using your laptop before powering on the FPM system:

1. **Preparation:** Obtain a clean microSD card (at least 16GB) and an SD card reader. Download and install the official **Raspberry Pi Imager** software on your laptop.
2. **Storage Connection:** Insert the microSD card into the SD card reader, and connect the reader to your laptop. *Note: Do not connect the Raspberry Pi itself to your laptop.*
3. **Configuring the Imager:** Open the Raspberry Pi Imager.
 - Under **Choose Device**, select your specific Raspberry Pi model.
 - Under **Choose OS**, select **Raspberry Pi OS (32-bit or 64-bit)**.
 - Under **Choose Storage**, select the connected microSD card.
4. **OS Customization (Crucial Step):** Click **Next**. When prompted to apply OS customization settings, click **Edit Settings**.
 - **General Tab:** Set a hostname (e.g., `fpm-pi`). Set a username and a secure password. Configure the wireless LAN by entering your local Wi-Fi network name (SSID) and password. Set your local time zone and keyboard layout.
 - **Services Tab:** Check the box to **Enable SSH** and select “Use password authentication.” This is mandatory for controlling the Pi remotely without a dedicated monitor.

Click **Save** to apply these configurations.

5. **Flashing the MicroSD Card:** Click **Yes** to confirm the OS customization. A warning will appear stating that all existing data on the SD card will be erased; click **Yes** to proceed. Wait for the software to write and verify the OS.
6. **First Boot:** Once the software says “Write Successful,” safely eject the microSD card from your laptop. Insert the flashed microSD card into the slot on the underside of the Raspberry Pi.

2.2 System Boot and Verification

1. **Power On:** Connect the Raspberry Pi power supply. Allow the system to boot into the operating system.
2. **Interface Verification:** Open a terminal and ensure both the Camera and SPI interfaces are active. You can check this by running `vcgencmd get_camera` to confirm the camera module is detected (should return `supported=1 detected=1`).

2.3 Optical Calibration

1. **Sample Stage Prep:** Place a transparent calibration slide (such as a USAF 1951 resolution test chart) on the stage area directly between the LED matrix and the camera lens.
2. **Center Illumination:** Run a test script to illuminate only the central LED (coordinate 8,8 on the 16x16 grid).
3. **Focusing:** View the live camera feed. Gently adjust the focus ring on the camera module until the features on the calibration slide are as sharp as possible. Use the allen key to adjust the focus.
4. **Centering Check:** Verify that the primary beam from the central LED strikes the center of the camera sensor.

2.4 Magnification Calibration

System magnification was empirically determined using a Thorlabs R1L3S5 stage micrometer (10 mm total length, 50 μm minor divisions). An image of the calibration ruler was captured, and ImageJ software was utilized to measure a known 1 mm interval spanning from the start of one major division to the next.

The measured distance in the image was determined to be 948.019 pixels. Given the Raspberry Pi Camera v2 sensor pixel size of 1.12 μm , the system's image scale and overall optical magnification were calculated as follows:

Step 1: Calculate the Image Scale

$$Scale = \frac{ActualDistance}{MeasuredPixels} = \frac{1000 \mu\text{m}}{948.019 \text{ pixels}} = 1.0548 \mu\text{m/pixel}$$

Step 2: Calculate Magnification (M)

$$M = \frac{SensorPixelSize}{ImageScale} = \frac{1.12 \mu\text{m}}{1.0548 \mu\text{m/pixel}} = 1.0618$$

Calibration Summary

Parameter	Value
Measured Length	948.019 pixels
Actual Distance	1000 μm (1 mm)
Calculated Scale	1.055 $\mu\text{m/pixel}$
Sensor Pixel Size	1.12 μm
System Magnification	1.06$\times$

Note: The validity of the 1.06 $\times$ magnification depends strictly on confirming that the measurement line drawn in ImageJ spans exactly one 1 mm interval on the Thorlabs calibration ruler. If the physical distance between the measured marks varies, the magnification value will change accordingly.

2.5 Objective Numerical Aperture (NA) Calibration

To optimize the FPM reconstruction algorithm, the usable objective Numerical Aperture (NA) of the Raspberry Pi Camera V2 lens was measured objectively. This was achieved by tracking the brightfield-to-darkfield transition of the LED array.

1. Data Acquisition A clear transparent slide was placed in focus. The LED matrix sequentially illuminated a center-out cross pattern, starting at the center LED (coordinate 7,7) and moving incrementally right, left, down, and up. To ensure a valid comparison of light intensity, camera settings (exposure time, frame duration, analogue gain) and LED brightness were strictly locked. Three repeated images were captured per LED position, which were later combined using a median average to eliminate random camera noise.

2. Algorithmic Boundary Detection A Region of Interest (ROI) was drawn over a clear transparent area in the image generated by the center LED. An automated script then measured the median brightness of this exact ROI for every subsequent outward LED. The boundary of the objective lens was defined as the point where the ROI's measured brightness fell below 50% of the initial center-LED brightness.

3. Mathematical NA Conversion Once the script identified the last "bright" LED and the first "dark" LED along a directional ray, the physical LED coordinates were converted into an illumination NA value. The system utilizes an LED pitch of 3.3 mm and a distance from the LED array to the sample of 61.8 mm.

The radial distance (r) of any LED from the center is calculated using the Pythagorean theorem based on the LED pitch. The specific illumination NA is then derived using the formula:

$$NA = \frac{r}{\sqrt{r^2 + d^2}}$$

where d is the fixed LED-to-sample distance (61.8 mm).

The final usable objective NA is calculated by mathematically interpolating the exact boundary between the calculated NA of the last bright LED and the first dark LED, and then taking the median boundary value across all four tested directional rays.

3 Phase 3: Data Acquisition (Operations)

3.1 Data Pre-Processing Pipeline

Prior to initiating the computationally heavy phase-retrieval reconstruction algorithms (such as EPRY), the raw DNG datasets must undergo several critical pre-processing steps. This ensures data integrity, minimizes artifacts, and maps the computational model accurately to the physical hardware.

- **LED Geometry Calibration:** The exact central LED and axis orientation might slightly deviate between physical assembly and the software grid. To resolve this, the system runs a short automated diagnostic search across candidate center LEDs and axis orientations. It selects the geometry configuration that yields the highest raw consistency score and valid pupil structure, ensuring the illumination angles are accurately mapped in the Fourier domain.
- **LED Exposure Normalization:** The acquisition workflow utilizes a variable, multi-shell exposure schedule (e.g., outer darkfield LEDs require significantly longer

exposure times than the brightfield center LEDs). To make the intensity values across all images mathematically comparable, each image is normalized by dividing its intensity by its relative exposure factor (calculated via exposure time and analog gain). An exposure gain cap is also applied to prevent the disproportionate amplification of sensor noise from the darkest edge images.

- **Dark Subtraction & Background Removal:** To mitigate baseline sensor noise, read noise, and hot pixels, dark subtraction is applied. By subtracting a reference dark frame (or calculating a border-based background floor) from the raw captures, the baseline noise is removed. This step is particularly critical for darkfield images, where the high-frequency scattering signals are weak and easily corrupted by noise.

3.2 Standard Scanning Workflow

To execute a complete FPM scan, adhere to the following sequence:

1. **Specimen Loading:** Carefully remove the calibration slide and place your target specimen on the stage.
2. **Environment Control:** Complete darkness is required to prevent stray light from polluting the low-intensity diffraction images. Turn off all room lights or cover the FPM setup with a light-blocking enclosure.
3. **Sequence Execution:** Launch the FPM Python automation script via the terminal. The sequence will autonomously:
 - Iterate through the defined LED grid (typically a spiral pattern radiating outward from the center).
 - Illuminate a single LED.
 - Trigger the camera to capture an uncompressed/RAW image.
 - Save the image to the designated output folder with its corresponding positional index.
 - Turn off the LED and proceed to the next position.
4. **Data Offload:** Once the script completes (generating up to 256 images depending on the configured grid), transfer the dataset via SSH, FTP, or physical USB drive to the processing workstation.

3.3 RGB FPM Reconstruction & Aligned Color Fusion

For setups utilizing sequential monochromatic LED illumination (capturing independent datasets for red, green, and blue light), the three colors must **not** be mixed inside a single FPM inverse problem. Standard FPM reconstruction algorithms strictly assume a single illumination wavelength per exposure dataset.

To achieve a full-color, high-resolution reconstruction, the system employs an independent recovery and aligned fusion pipeline:

1. **Red Channel Recovery:** The dataset captured under red LED illumination is extracted from the demosaiced red sensor channel. The phase-retrieval algorithm is run utilizing the specific red LED wavelength parameter (e.g., 0.630 μm) to synthesize the high-resolution red amplitude image.

2. **Green Channel Recovery:** The green dataset is extracted from the green sensor channel. The algorithm is executed independently using the green LED wavelength (e.g., 0.530 μm) to synthesize the green amplitude image.
3. **Blue Channel Recovery:** The blue dataset is extracted from the blue sensor channel. The algorithm is executed independently using the blue LED wavelength (e.g., 0.470 μm) to synthesize the blue amplitude image.
4. **Color Fusion:** Because different wavelengths diffract differently, the three resulting high-resolution amplitude images will have slight variations in magnification

4 Phase 4: Fast Balanced Computational Reconstruction (DR/EPRY)

To generate the final high-resolution image across the complete camera Field of View (FOV), this FPM setup utilizes a fast, balanced reconstruction pipeline. This approach combines Directed Relaxation (DR) and the Embedded Pupil Recovery (EPRY) algorithm, optimized for speed and memory efficiency through disk caching.

4.1 RGB Channel Separation and Preprocessing

For sequential monochromatic RGB acquisition, the three color channels must be processed independently. To remove the primary full-FOV bottleneck, this algorithm preprocesses each raw DNG file only once per color:

- (a) **Color Sequence:** Reconstruct the channels in the following order: RED ($\approx 0.630 \mu\text{m}$), GREEN ($\approx 0.530 \mu\text{m}$), and BLUE ($\approx 0.470 \mu\text{m}$).
- (b) **Single-Pass Preprocessing:** Each DNG is demosaiced, dark-subtracted, and normalized by its actual exposure and analog gain *once*.
- (c) **Disk Caching:** The preprocessed full-frame arrays for the selected LEDs are cached to disk. Subsequent reconstruction tiles are read directly from this memory-mapped cache, drastically reducing computational overhead.

4.2 Balanced Reconstruction Parameters

This pipeline is configured for a "balanced" run—prioritizing a practical, complete FOV test before committing to a maximum-quality, computationally expensive run.

The default balanced settings are:

- **Tile Size:** 512×512 pixels.
- **Tile Overlap:** 16 pixels.
- **Illumination Shells:** Shells 0-6 (utilizing 196 LEDs) instead of the full 256.
- **EPRY Passes:** 2 pupil stages with 4 iterations per stage.

- **Backend:** Hardware acceleration is utilized via a CuPy GPU backend if available; otherwise, it falls back to multi-core SciPy CPU processing.

Note: For a final, maximum-quality scientific run, these parameters should be increased (e.g., using all 7 shells, 3 EPRY stages, and 6 iterations per stage).

4.3 The DR/EPRY Update Logic

The core reconstruction operates via regularized gradient-descent, using the identical DR and EPRY update equations established in the theoretical model.

1. The Pupil Update (Denominator $|P|^2 + \alpha$)

$$O_{update} = step_size \times \left(\frac{1}{\max |P|} \right) \times |P| \times conj(P) \times \frac{d\psi}{|P|^2 + \alpha}$$

2. The Object Update (Denominator $|O_{crop}|^2 + \beta$)

$$P_{update} = \left(\frac{1}{O_{\max}} \right) \times |O_{crop}| \times conj(O_{crop}) \times \frac{d\psi}{|O_{crop}|^2 + \beta} \times P_s$$

Where $d\psi$ is the difference in the pupil-plane spectrum between the Fourier-magnitude-projected exit wave and the current estimate.

4.4 Sub-Pixel LED Position Correction

Because the illumination angle of each LED is calculated from a theoretical grid, its (k_x, k_y) position in the object spectrum rarely lands exactly on an integer pixel. To prevent the spatially-coherent position errors that cause periodic "grid" artifacts, a sub-pixel shift correction is applied locally via the Fourier shift theorem:

$$O_{crop_subpixel} = FT \left(IFT(O_{crop}) \times e^{i \cdot 2\pi \cdot (f_x \cdot \frac{x}{m} + f_y \cdot \frac{y}{m})} \right)$$

This residual fractional phase ramp is precomputed once per color to maintain the speed of the fast-path algorithm.

4.5 Interpreting Raw Results & Target Validation

Before running computationally heavy reconstruction algorithms or transitioning to complex biological samples, verify the integrity of the raw dataset and the system's spatial resolution using a calibration target.

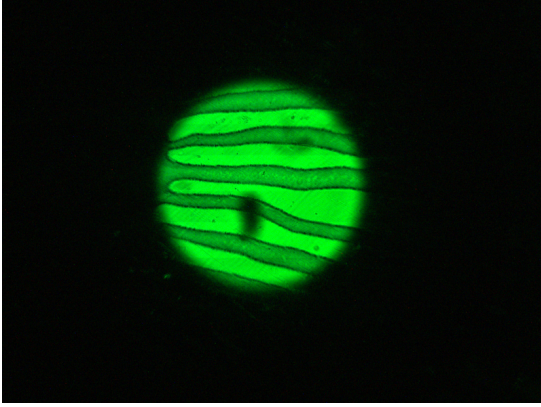
1. Specimen Selection Protocol:

- **Initial Calibration Phase:** *Do not use biological samples during the initial setup and algorithm testing.* Biological specimens introduce variable contrast, unknown thickness, and scattering ambiguities that make it incredibly difficult to isolate hardware or algorithmic errors.

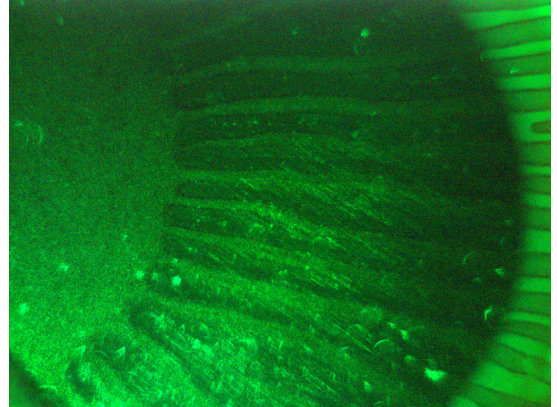
- **Resolution Targets:** Begin with a standard, high-contrast calibration slide, such as a Thorlabs R1L3S5 Ronchi ruling target. This target includes specific line pairs, grids, interdigitated lines, and concentric circles. In this validation workflow, the system specifically targeted the resolution of fine high-frequency structures, successfully tracking and resolving down to the **81 LP/mm** (line pairs per millimeter) line to definitively confirm synthetic aperture expansion.

2. Raw Image Verification:

- **Brightfield Images (Center LEDs):** These should look like standard, low-resolution microscope images with high overall intensity, cleanly displaying the macro-features of the test target.
- **Darkfield Images (Outer/Edge LEDs):** These should be significantly darker, highlighting only the sharp edges, boundaries, and high-frequency grating lines of the specimen. If these images are totally black or uniform, the exposure time in your capture script may need to be increased.



Brightfield Image



Darkfield Image

Figure 2: Comparison of raw sensor captures from center LED with green illumination versus edge LED illumination.

4.6 Evaluating Reconstructed Results

Following the fast balanced computational reconstruction (DR/EPRY), the dataset yields high-resolution amplitude and phase arrays. Because the system acquires data sequentially, the reconstruction is processed independently for each color channel before being aligned and fused into the final image.

1. Target Reconstruction Analysis (Raw, Amplitude, and Phase): It is crucial to first perform reconstruction on a standard calibration target—specifically the Thorlabs R1L3S5 Ronchi ruling used in this setup—before transitioning to complex biological specimens. Biological samples introduce variable contrast, unknown thicknesses, and complex scattering properties that make isolating hardware misalignment or algorithmic errors extremely difficult. By validating with the R1L3S5 target, the multi-angle dataset yields reliable, quantitative high-resolution maps. Because FPM retrieves both the complex amplitude and phase of the sample, the

output provides critical insight into transparent, unstained periodic structures that are typically invisible in standard brightfield microscopy.

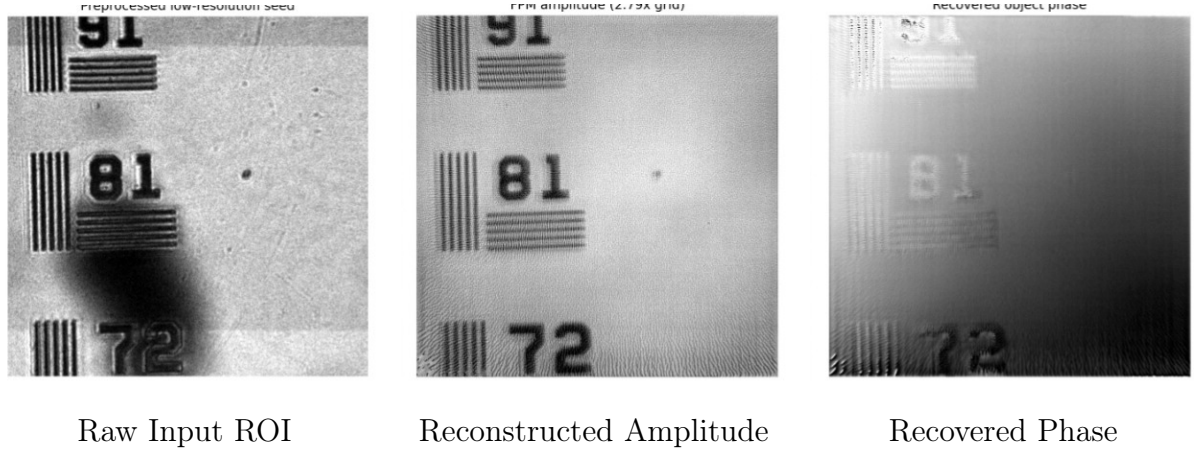


Figure 3: Side-by-side diagnostic views showing the raw low-resolution capture (left), the high-resolution amplitude resolving the 81 LP/mm grating (center), and the quantitative optical phase delay map (right).

2. Individual Color Channels: The reconstructions for the RED, GREEN, and BLUE channels should each demonstrate enhanced resolution, improved contrast, and clear structural details corresponding to their specific illumination wavelengths.

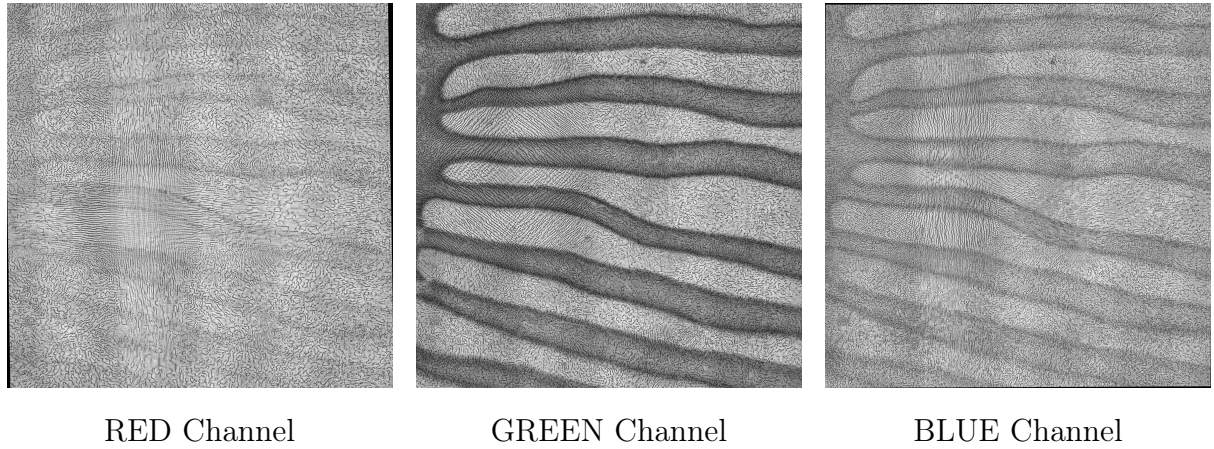
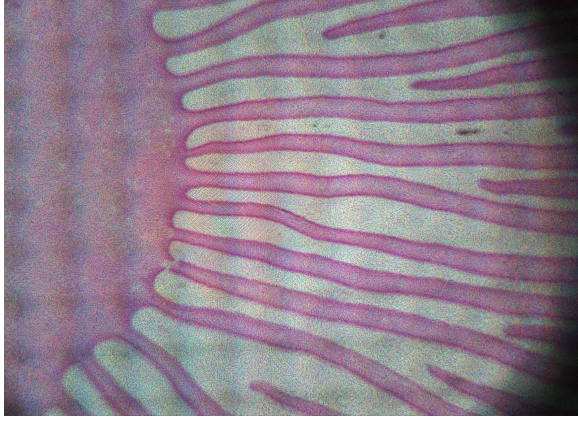
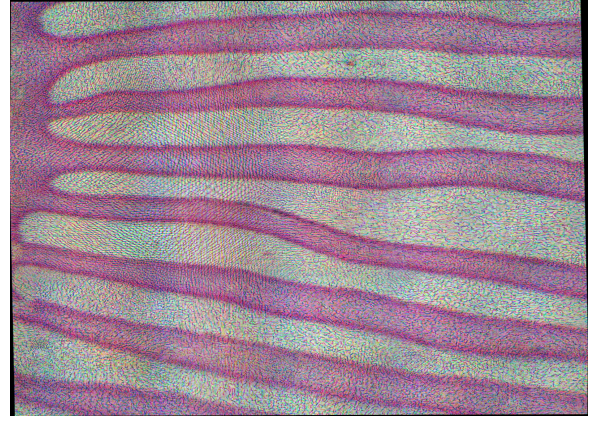


Figure 4: Independent high-resolution amplitude reconstructions for the Red ($\approx 0.630 \mu\text{m}$), Green ($\approx 0.530 \mu\text{m}$), and Blue ($\approx 0.470 \mu\text{m}$) channels prior to color fusion.

3. Full Field-of-View (FOV) and High-Resolution ROI: Once the independent channels are stitched and registered, they form the final composite image. The Full FOV image provides a wide macroscopic overview of the specimen, overcoming the traditional limits of standard microscope objectives. The cropped 512×512 Region of Interest (ROI) highlights the sub-micron resolution achieved computationally.



Full FOV Reconstruction



Cropped 512×512 ROI

Figure 5: Final synthesized results displaying the complete, wide-field macroscopic view (left) alongside a highly detailed microscopic Region of Interest crop (right).

5 Phase 4: Troubleshooting & References

5.1 Troubleshooting Common Operational Issues

Symptom	Corrective Action
Camera fails to initialize	Soft-reboot the Pi. If the issue persists, carefully check that the CSI ribbon cable has not been accidentally dislodged from the Pi's port during handling.
Reconstruction algorithm produces artifacts	Artifacts usually stem from stray ambient light during capture or sub-pixel movement of the sample. Ensure the enclosure is perfectly dark and the table is isolated from vibrations.
Darkfield images are too noisy/dim	Increase the exposure time parameter in your capture script specifically for the outer rings of the LED matrix.
LED matrix does not light up	Verify that the SPI interface is enabled via <code>sudo raspi-config</code> and that the Python script has the necessary permissions (<code>sudo</code> may be required for GPIO access).

5.2 Code Repository

The reconstruction code required to process the acquired datasets is available on GitHub at <https://github.com/johnmeshreki/FPM>.

5.3 Literature and Assembly References

This setup and operational methodology are based on several key works in the field of computational imaging:

- "Low-cost, sub-micron resolution, wide-field computational microscopy using open-source hardware" by Aidukas et. al.
- "Miniature Fourier ptychography microscope using Raspberry Pi camera and hardware" by Konda et. al.
- "Low-Cost Fourier Ptychography Microscope Kit" by Guzmán et. al.

For further details regarding the physical assembly of the optics and chassis, reference the supplementary material provided at <https://www.nature.com/articles/s41598-019-43845-9> and the presentation available at <https://librehub.github.io/2022/07/14/esteban-vera.html>.